

Insurance Knowledge Base

Comprehensive RAG Data Ingestion Document

This document serves as the primary knowledge corpus for the Insurance Claim Support AI Agent built with LangMem and RAG. It contains detailed information across all major insurance domains including health, auto, life, property, liability insurance — covering policies, claim procedures, terminology, regulatory frameworks, and decision-support data.

Intended Use: Data ingestion into a vector database (e.g., FAISS, Chroma, Pinecone) for Retrieval-Augmented Generation (RAG) pipelines powering an AI copilot for insurance claims support.

Category	Coverage
Insurance Types	Health, Auto, Life, Property, Liability, Travel
Claims Process	FNOL, Investigation, Settlement, Appeals
Policy Terms	Definitions, Exclusions, Endorsements
Regulatory	State/Federal Laws, Compliance, HIPAA
Decision Support	Coverage Matrix, Fraud Indicators, SLAs

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Chapter 1: Fundamentals of Insurance

1.1 What Is Insurance?

Insurance is a legal contract (policy) in which an insurer (the insurance company) promises to compensate an insured (policyholder) for financial losses arising from specific covered events in exchange for periodic payments called premiums. Insurance is a risk-transfer mechanism that allows individuals and businesses to protect themselves against unpredictable financial losses.

The fundamental principle underlying all insurance is risk pooling: many people pay into a common fund, and those who suffer losses are compensated from that fund. This spreads risk across a large population, making individual losses manageable.

1.2 Core Insurance Principles

Principle of Indemnity

The insured should not profit from a loss. Insurance aims to restore the insured to their financial position before the loss occurred — no more, no less. This prevents moral hazard and over-insurance.

Principle of Insurable Interest

The policyholder must have a financial stake in the insured subject matter. For example, you can insure your own car but not a neighbour's car because you would not suffer a direct financial loss if your neighbour's car was damaged.

Principle of Utmost Good Faith (Uberrimae Fidei)

Both parties — insurer and insured — must disclose all material facts honestly. Concealment or misrepresentation of material information can void the policy.

Principle of Subrogation

After compensating the insured, the insurer acquires the right to pursue third parties responsible for the loss. For example, if another driver causes an accident and your insurer pays your claim, your insurer can then sue the at-fault driver to recover that payment.

Principle of Contribution

If multiple insurance policies cover the same risk, each insurer contributes proportionally to the loss. This prevents the insured from claiming the full amount from multiple insurers.

1.3 Types of Insurance — Overview

Insurance Type	What It Covers	Key Benefit
Health	Medical expenses, hospitalization, prescriptions	Protects against high healthcare costs
Auto	Vehicle damage, liability, bodily injury	Legal requirement in most jurisdictions
Life	Death benefit to beneficiaries	Income replacement, estate planning
Homeowners	Dwelling, personal property, liability	Protects largest personal asset
Renters	Personal property, liability (no dwelling)	Affordable protection for tenants
Liability	Third-party injury/property damage claims	Business and personal asset protection
Travel	Trip cancellation, medical evacuation	Peace of mind while traveling
Disability	Income replacement if unable to work	Protects earning capacity

1.4 The Insurance Policy Structure

Every insurance policy contains the following standard components:

- **Declarations Page (Dec Page):** Summary of coverage — policyholder name, policy number, effective dates, coverage limits, deductibles, and premium amount.
- **Insuring Agreement:** The insurer's promise to pay covered losses. Defines the scope of coverage.
- **Conditions:** Obligations the policyholder must fulfill to maintain coverage (e.g., timely premium payment, prompt loss notification, cooperation with investigation).
- **Exclusions:** Specific events, perils, or circumstances NOT covered by the policy. Common exclusions include intentional acts, acts of war, and wear and tear.
- **Endorsements / Riders:** Amendments that modify, add, or restrict coverage beyond the standard policy.
- **Definitions:** Precise meanings of terms used throughout the policy to prevent ambiguity.

1.5 Premium Calculation Factors

Insurers use actuarial science and underwriting to price policies. Key factors include:

Factor	Example Impact
Age	Older drivers or individuals may pay higher premiums
Claims History	Prior claims indicate higher risk; premiums increase
Credit Score	Lower scores correlate with higher claims frequency
Location/ZIP Code	Urban areas with higher theft/accident rates cost more

Factor	Example Impact
Coverage Limits	Higher limits mean higher premiums
Deductible Amount	Higher deductible = lower premium, more out-of-pocket risk
Occupation	High-risk occupations increase life/disability premiums
Gender & Marital Status	Statistically correlated with risk levels

Chapter 2: Health Insurance

2.1 Overview of Health Insurance

Health insurance covers the cost of medical and surgical expenses incurred by the insured. Depending on the plan type, the insurer may pay providers directly or reimburse the insured after they pay out-of-pocket. Health insurance is critical in countries where healthcare costs are high, such as the United States.

2.2 Types of Health Insurance Plans

HMO — Health Maintenance Organization

Members must select a Primary Care Physician (PCP) who coordinates all care and provides referrals to specialists. Coverage is limited to in-network providers except in emergencies. HMOs generally have lower premiums and out-of-pocket costs but less flexibility.

PPO — Preferred Provider Organization

Members can see any doctor without a referral. Out-of-network care is covered but at a higher cost. PPOs offer more flexibility than HMOs but typically carry higher premiums.

EPO — Exclusive Provider Organization

Similar to HMOs in that members must use in-network providers, but no PCP referral is required for specialist visits. Out-of-network care is generally not covered except in emergencies.

HDHP — High Deductible Health Plan

Plans with deductibles of \$1,600+ for individuals (2024 IRS threshold). Often paired with a Health Savings Account (HSA). Lower premiums make HDHPs attractive for healthy individuals.

POS — Point of Service

A hybrid of HMO and PPO plans. Members choose a PCP but can see out-of-network providers at higher cost. Referrals are required for specialist visits.

2.3 Key Health Insurance Terms

Term	Definition
Premium	Monthly payment to maintain coverage regardless of usage
Deductible	Amount paid out-of-pocket before insurance begins paying

Term	Definition
Copay	Fixed amount paid at time of service (e.g., \$30 per visit)
Coinsurance	Percentage of costs shared after deductible (e.g., 80/20 split)
Out-of-Pocket Maximum	Annual cap on what insured pays; insurer covers 100% after
Network	Group of providers contracted with the insurer at negotiated rates
Pre-authorization	Insurer approval required before certain procedures
EOB	Explanation of Benefits — statement showing how claim was processed
Formulary	List of prescription drugs covered by the plan
COBRA	Continuation coverage for those who lose employer-sponsored insurance

2.4 What Health Insurance Typically Covers

- Preventive care: annual physicals, vaccinations, screenings (often at 100% in-network)
- Emergency services: ER visits, ambulance transportation
- Hospitalization: inpatient care, surgery, nursing services
- Maternity and newborn care
- Mental health and substance use disorder services
- Prescription drugs (subject to formulary and tier structure)
- Rehabilitation services: physical therapy, occupational therapy
- Pediatric services including vision and dental for children

2.5 Common Health Insurance Exclusions

- Elective/cosmetic procedures not medically necessary
- Experimental or investigational treatments
- Long-term care and custodial care
- Dental and vision for adults (unless separately purchased)
- Weight-loss surgery (unless meeting specific criteria)
- Out-of-network care (for HMO/EPO plans)

2.6 Health Insurance Claims Process

When a covered individual receives medical care, the following process typically occurs:

Step 1 — Receive Service: Patient visits a provider and presents insurance card. Provider verifies eligibility in real time.

Step 2 — Provider Bills Insurer: Provider submits a claim (CMS-1500 or UB-04 form) using standardized medical codes (ICD-10, CPT, HCPCS).

Step 3 — Claims Adjudication: Insurer reviews claim for eligibility, coverage, medical necessity, and coding accuracy. The claim is approved, denied, or pended for additional information.

Step 4 — Explanation of Benefits (EOB): Insurer sends EOB to both provider and patient detailing what was billed, what was allowed, what the plan paid, and what the patient owes.

Step 5 — Patient Responsibility: Patient pays any remaining balance (copay, deductible, coinsurance) directly to provider.

2.7 Health Insurance Appeals

Denied claims can be appealed through the insurer's internal process and, if needed, through an Independent Medical Review (IMR) or external appeal. Key timelines:

Appeal Type	Timeframe
Urgent/Concurrent Care Internal Appeal	Within 72 hours of denial
Standard Pre-Service Internal Appeal	Within 30 days
Post-Service Internal Appeal	Within 60 days of denial
External Review Request	Within 4 months of final internal denial

Note: ACA-compliant plans must follow federal appeal regulations. State laws may provide additional protections.

Chapter 3: Auto Insurance

3.1 Overview

Auto insurance provides financial protection against physical damage or bodily injury resulting from traffic collisions and against liability that could also arise from incidents in a vehicle. Auto insurance is mandatory in 49 U.S. states (New Hampshire requires proof of financial responsibility instead).

3.2 Types of Auto Insurance Coverage

Liability Coverage

Covers bodily injury and property damage the insured causes to others in an at-fault accident. Split into:

- Bodily Injury Liability (BI): Covers medical expenses, lost wages, and pain and suffering for injured third parties.
- Property Damage Liability (PD): Covers damage to others' vehicles or property.
- Expressed as a split limit (e.g., 100/300/100 = \$100K per person / \$300K per occurrence / \$100K property damage) or combined single limit.

Collision Coverage

Pays for damage to the insured's vehicle resulting from a collision with another vehicle or object, regardless of fault. Subject to a deductible.

Comprehensive Coverage

Covers non-collision damage including theft, vandalism, fire, flooding, hail, falling objects, and animal strikes. Also subject to a deductible.

Uninsured/Underinsured Motorist (UM/UIM)

Protects the insured when the at-fault driver has no insurance (UM) or insufficient coverage to pay for all damages (UIM).

Medical Payments (MedPay) / Personal Injury Protection (PIP)

Covers medical expenses for the insured and passengers regardless of fault. PIP (required in no-fault states) also covers lost wages and other expenses.

Rental Reimbursement

Covers cost of a rental car while the insured's vehicle is being repaired after a covered loss.

Roadside Assistance / Towing

Covers towing, flat tire changes, jump starts, lockout services, and fuel delivery.

3.3 Auto Insurance Rating Factors

Rating Factor	Impact on Premium
Driving Record	Clean record = lowest rate; DUI/DWI can double or triple premium
Age	Teen drivers (16-25) and seniors pay more; lowest rates 30-65
Vehicle Make/Model	Sports cars, luxury vehicles cost more to insure
Annual Mileage	More miles = more exposure = higher premium
Credit Score	Allowed in most states; poor credit raises premiums significantly
Location	Dense urban areas with higher theft/accident rates cost more
Coverage Limits	Higher limits increase premium proportionally
Deductible	Higher deductibles reduce comprehensive/collision premiums
Discounts	Multi-car, safe driver, good student, bundling reduce premiums

3.4 Auto Insurance Claims Process

When an accident occurs, the insured should take the following steps:

At the Scene: Ensure safety, call 911 if injuries exist, exchange contact and insurance information with all parties, photograph the scene and damage, collect witness information, file a police report.

Within 24-48 Hours: Notify your insurer by phone or app. Provide accident details, policy number, and contact information for all parties involved.

Claims Assignment: Insurer assigns a claims adjuster who contacts all parties, requests recorded statements, and schedules vehicle inspection.

Vehicle Damage Evaluation: Adjuster or appraiser inspects vehicle and creates a repair estimate. Insured may use insurer's direct repair network or choose their own shop.

Settlement: Insurer pays repair shop directly (minus deductible) or issues payment to insured. For total losses, insurer pays Actual Cash Value (ACV) of the vehicle.

3.5 Total Loss Determination

A vehicle is declared a total loss when the cost of repairs exceeds a threshold relative to the vehicle's Actual Cash Value (ACV). The threshold varies by state but is commonly 70-80% of ACV.

ACV = Replacement Cost - Depreciation. Insurers use market data (Kelley Blue Book, Black Book, CCC ONE) to determine ACV.

Total Loss Settlement Options

- Accept ACV payout and surrender title to insurer
- Retain salvage vehicle (ACV paid minus salvage value)
- Gap Insurance covers the difference between ACV and remaining loan balance

3.6 Diminished Value Claims

After a repaired vehicle's market value is lower than a similar vehicle that has never been in an accident, the insured may be entitled to a Diminished Value (DV) claim against the at-fault party's insurer. Three types exist: Inherent DV (most common), Repair-Related DV, and Instant DV.

3.7 State Minimum Requirements (Sample)

State	Minimum BI Limit	Minimum PD Limit	PIP Required
California	\$15,000/\$30,000	\$5,000	No
Texas	\$30,000/\$60,000	\$25,000	No
Florida	\$10,000 (PIP)	\$10,000	Yes
New York	\$25,000/\$50,000	\$10,000	Yes
Michigan	Unlimited (default)	\$1,000,000	Yes

Note: Minimum coverage often insufficient. Industry recommendation: 100/300/100 with UM/UIM.

Chapter 4: Life Insurance

4.1 Overview

Life insurance provides a financial benefit (death benefit) to beneficiaries upon the death of the insured. It is used for income replacement, debt coverage, estate planning, business continuity, and charitable giving. The two primary categories are term life insurance and permanent life insurance.

4.2 Types of Life Insurance

Term Life Insurance

Provides coverage for a specified period (10, 20, or 30 years). If the insured dies during the term, beneficiaries receive the death benefit. If the policy expires while the insured is alive, there is no payout. Term life is the most affordable option and is ideal for covering specific financial obligations like a mortgage or raising children.

Whole Life Insurance

A type of permanent life insurance providing lifelong coverage with a guaranteed death benefit. Premiums are fixed, and the policy builds cash value at a guaranteed rate that can be borrowed against or withdrawn. Premiums are significantly higher than term life.

Universal Life Insurance (UL)

Permanent coverage with flexible premiums and death benefits. Includes a cash value component that earns interest based on current market rates (with a minimum guaranteed rate). Variants include Indexed Universal Life (IUL — linked to stock market indices) and Variable Universal Life (VUL — invested in sub-accounts).

Variable Life Insurance

Death benefit and cash value fluctuate based on investment performance of chosen sub-accounts (similar to mutual funds). Offers highest growth potential but also highest risk — cash value can decline.

4.3 Life Insurance Key Terms

Term	Definition
Death Benefit	Amount paid to beneficiaries upon insured's death
Beneficiary	Person(s) or entity designated to receive the death benefit

Term	Definition
Premium	Regular payment to keep the policy in force
Cash Value	Savings component in permanent policies that grows over time
Surrender Value	Amount received if policyholder cancels the policy
Rider	Optional add-on coverage (e.g., waiver of premium, accidental death)
Grace Period	30-day window to pay overdue premium before policy lapses
Contestability Period	First 2 years insurer may deny claims for misrepresentation
Incontestability Clause	After 2 years, policy cannot be voided for misrepresentation
Policy Loan	Loan against cash value; does not require repayment but reduces benefit if unpaid

4.4 Life Insurance Underwriting

Underwriting determines the risk profile of the applicant and sets the premium. Methods include:

- **Fully Underwritten:** Requires medical exam, blood work, urinalysis, EKG (for high coverage amounts), and detailed health history questionnaire. Provides lowest premiums for healthy applicants.
- **Simplified Issue:** No medical exam; approval based on health questionnaire. Higher premiums; typically lower coverage limits (\$500K or less).
- **Guaranteed Issue:** No health questions or exam. Highest premiums; lower coverage. Usually offered to those aged 50-85 for final expense coverage. Graded death benefit for first 2 years.

4.5 Beneficiary Designations

Proper beneficiary designation is critical. Types of beneficiaries:

- **Primary Beneficiary:** First in line to receive the death benefit.
- **Contingent Beneficiary:** Receives benefit if primary predeceases the insured.
- **Revocable Beneficiary:** Can be changed at any time by policyholder.
- **Irrevocable Beneficiary:** Cannot be changed without their consent. Common in divorce settlements.

Note: Beneficiary designations override a will. Always keep beneficiary information updated after major life events.

4.6 Common Life Insurance Riders

Rider	Description
Waiver of Premium	Waives premiums if insured becomes totally disabled
Accidental Death Benefit	Pays additional benefit (often 2x) for accidental death
Child Term Rider	Provides term coverage for insured's children at low cost
Accelerated Death Benefit	Allows access to portion of benefit if terminally ill
Long-Term Care Rider	Pays LTC expenses if insured cannot perform 2 of 6 ADLs
Return of Premium	Refunds all premiums if insured outlives the term
Guaranteed Insurability	Right to buy additional coverage without evidence of insurability

4.7 Life Insurance Claims

Filing a life insurance claim requires: certified copy of the death certificate, completed claim form (claimant's statement), and the original policy document if available. Claims are generally paid within 30-60 days of receiving all required documentation. Contestable claims (within 2 years of policy issue) take longer as the insurer reviews the application for misrepresentation.

Common reasons for claim delays or denials: death during contestability period, policy lapse due to non-payment, suicide within exclusion period (typically 2 years), material misrepresentation on the application, death from excluded activity (e.g., aviation, war).

Chapter 5: Property & Homeowners Insurance

5.1 Overview

Homeowners insurance (HO) protects the dwelling, other structures on the property, personal property, and provides liability coverage. It is typically required by mortgage lenders. Renters insurance (HO-4) covers personal property and liability for tenants but not the dwelling itself.

5.2 Homeowners Policy Forms

Form	Type	Coverage Scope
HO-1	Basic Form	Named perils only; very limited; rarely sold
HO-2	Broad Form	16 named perils; more common than HO-1
HO-3	Special Form	Open perils for dwelling; named perils for personal property; most common
HO-4	Renters Insurance	Personal property + liability; no dwelling coverage
HO-5	Comprehensive Form	Open perils for both dwelling and personal property
HO-6	Condo Insurance	Interior dwelling + personal property; no exterior structure
HO-8	Older/Historic Homes	Actual cash value; modified coverage for homes hard to replace

5.3 Standard Coverage Components

Coverage A — Dwelling

Covers the physical structure of the home including walls, roof, floors, built-in appliances, and attached structures like a garage. Coverage should equal the replacement cost of the home (not market value).

Coverage B — Other Structures

Covers detached structures on the property: detached garages, fences, sheds, guest houses. Typically 10% of Coverage A limits.

Coverage C — Personal Property

Covers belongings inside and outside the home (e.g., belongings stolen from a car). Coverage is typically 50-70% of Coverage A. High-value items (jewelry, art, firearms, electronics) may have sub-limits and require scheduled endorsements.

Coverage D — Loss of Use / Additional Living Expenses (ALE)

Covers extra living costs if the home is uninhabitable due to a covered loss. Includes hotel, dining, laundry, and storage costs. Typically 20-30% of Coverage A with a time limit.

Coverage E — Personal Liability

Protects against lawsuits for bodily injury or property damage caused by the insured or family members. Standard limits are \$100K-\$300K.

Coverage F — Medical Payments to Others

Pays medical expenses for guests injured on the property regardless of fault. Does not apply to insured or residents. Limits typically \$1,000-\$5,000.

5.4 Named Perils vs. Open Perils

Named Perils: Only losses explicitly listed in the policy are covered. Common named perils include fire, lightning, windstorm, hail, explosion, riot, aircraft, vehicles, smoke, vandalism, theft, falling objects, weight of ice/snow, water damage from household systems, electrical surge.

Open Perils (All-Risk): All causes of loss are covered UNLESS specifically excluded. Broader protection. Exclusions typically include flood, earthquake, wear and tear, intentional acts, war, nuclear hazard.

5.5 Major Exclusions

- Flooding — requires separate NFIP or private flood policy
- Earthquakes — requires separate earthquake endorsement or policy
- Sewer/drain backup — available as an endorsement
- Mold — limited coverage; often only if result of a covered loss
- Landslide and earth movement
- Power failure (off premises)
- Intentional acts and criminal acts
- Neglect and deferred maintenance
- Business activities (home business endorsement available)

5.6 Property Claims Process

Step 1 — Mitigate Further Damage: Policyholder has a duty to prevent additional damage (e.g., tarp a damaged roof, extract standing water). Keep receipts for emergency repairs.

Step 2 — Document the Loss: Photograph all damage before cleanup. Create a detailed inventory of damaged/destroyed personal property with estimated values and purchase dates.

Step 3 — File the Claim: Notify insurer promptly (most policies require "timely" notice). Provide policy number, date of loss, cause of loss, and description of damage.

Step 4 — Adjuster Inspection: Insurer assigns a field adjuster or uses virtual inspection. Adjuster prepares a Scope of Loss and repair estimate.

Step 5 — Payment: Insurer issues ACV payment initially (if RCV policy). After repairs, insured submits receipts and insurer pays recoverable depreciation to reach full replacement cost.

5.7 Actual Cash Value vs. Replacement Cost Value

	Actual Cash Value (ACV)	Replacement Cost Value (RCV)
Definition	Replacement cost minus depreciation	Full cost to replace with similar item
Payout	Lower — accounts for age/wear	Higher — no depreciation deducted
Example	5yr old TV: RCV \$800 - Depreciation \$300 = \$500	5yr old TV: Full \$800 paid
Premium Impact	Lower premium	Higher premium (typically 10-15% more)
Best For	Budget-conscious; older items	Comprehensive protection

Chapter 6: Liability Insurance

6.1 Overview

Liability insurance protects the insured against claims of bodily injury, property damage, personal injury (defamation, false arrest), and advertising injury caused by the insured's actions, products, or operations. It pays defense costs and judgments up to policy limits.

6.2 Types of Liability Insurance

General Liability (CGL)

Commercial General Liability insurance is foundational for businesses. It covers bodily injury and property damage arising from business operations, products, and completed operations. Also includes personal and advertising injury (libel, slander, copyright infringement).

Professional Liability (E&O;)

Errors and Omissions insurance protects professionals (doctors, lawyers, accountants, consultants, real estate agents) against claims of negligence, errors, or failure to perform professional duties. Claims-made basis means coverage must be in force when the claim is filed, not just when the error occurred.

Directors & Officers (D&O;)

Protects corporate directors and officers against personal liability for decisions made in their official capacity. Covers securities claims, shareholder lawsuits, regulatory investigations, and employment practices claims.

Umbrella / Excess Liability

Provides additional limits above underlying policies (auto, homeowners, CGL). Umbrella also fills coverage gaps in underlying policies. Most individuals and businesses should consider \$1M-\$5M umbrella coverage.

Product Liability

Covers manufacturers, distributors, and retailers against claims that a product caused bodily injury or property damage. Covered defects include design defects, manufacturing defects, and failure to warn.

6.3 Claims-Made vs. Occurrence Policies

	Claims-Made	Occurrence
Trigger	Claim must be FILED during policy period	Injury/damage must OCCUR during policy period
Tail Coverage	Needed after policy cancellation (extended reporting period); coverage follows the event	Not needed; coverage follows the event
Prior Acts	Retroactive date defines how far back coverage extends	Does not extend
Common Use	Professional liability, D&O, cyber	General liability, auto, homeowners
Cost	Generally lower initially; rises with experience	Typically higher but more straightforward

6.4 Liability Coverage Limits

- **Per-Occurrence Limit:** Maximum paid for any single covered event or claim.
- **Aggregate Limit:** Maximum total paid for all claims during the policy period.
- **Per-Person Limit:** Maximum paid to any single injured party in a liability claim.
- **Split Limits:** Separate limits for bodily injury per person, per occurrence, and property damage.

6.5 Liability Claims Process

When a third party makes a liability claim against the insured:

- 1. Tender the Claim:** Insured notifies insurer of any incident that could result in a claim. The insurer has a duty to defend (pays legal costs) and may have a duty to indemnify (pay damages).
- 2. Investigation:** Insurer investigates the incident, liability exposure, and damages. This includes reviewing incident reports, witness statements, and expert opinions.
- 3. Reservation of Rights:** Insurer may defend the insured while reserving the right to deny coverage if the claim is later found to fall outside policy terms.
- 4. Settlement or Litigation:** Insurer may settle with the claimant or defend through trial. The insured should not admit liability or settle without insurer consent.

Note: Never admit fault at the scene of an accident. Statements can be used against the insured in liability claims.

Chapter 7: Travel Insurance

7.1 Overview

Travel insurance provides financial protection against unexpected events that can disrupt travel plans and cause financial loss. It is highly recommended for international travel, cruises, adventure travel, and any trip with significant non-refundable expenses.

7.2 Core Travel Insurance Coverages

Coverage	What It Covers	Typical Limit
Trip Cancellation	Non-refundable expenses if trip cancelled for covered reason	100% of insured trip cost
Trip Interruption	Extra costs + unused portions if trip cut short	150% of insured trip cost
Trip Delay	Meals, lodging if trip delayed 6+ hours	\$150-\$300/day
Emergency Medical	Hospital, doctor, prescription costs abroad	\$50K - \$500K
Medical Evacuation	Emergency transport to nearest adequate facility	\$100K - \$1M
Baggage Loss	Reimbursement for lost, stolen, damaged luggage	\$1,500 - \$3,000
Baggage Delay	Emergency purchases if bags delayed 12+ hours	\$150 - \$500
Cancel for Any Reason	75% reimbursement regardless of reason for cancellation	75% of trip cost

7.3 Covered Reasons for Trip Cancellation

- Unexpected illness, injury, or death of insured, travel companion, or family member
- Natural disaster at destination or departure location
- Terrorism or mandatory evacuation
- Employer termination (involuntary layoff)
- Jury duty or court subpoena
- Home damage due to covered peril making home uninhabitable
- Bankruptcy of travel supplier (airline, cruise line)

7.4 What Travel Insurance Does NOT Cover

- Pre-existing conditions (unless waiver purchased within 10-21 days of initial deposit)
- Cancel for Any Reason: voluntary cancellations without a CFAR rider

- Extreme sports unless adventure rider purchased
- Travel to countries under State Department Do Not Travel advisories
- Losses due to intoxication or drug use
- Epidemics/pandemics (COVID-19 coverage now varies by policy)
- Change of mind

Chapter 8: Claims Processing & Procedures

8.1 First Notice of Loss (FNOL)

FNOL is the initial report made to an insurer following a loss. Best practices for an effective FNOL:

- Report promptly — most policies require 'timely' or 'prompt' notice; delays can jeopardize coverage
- Provide complete information: date, time, location, description of incident, parties involved, witnesses
- Do not admit fault or liability
- Obtain a claim number and adjuster contact information
- Document everything: photos, video, written records

8.2 Claim Lifecycle

Phase 1 — Receipt & Assignment: Claim received by insurer. Claim number assigned. File opened and adjuster assigned based on line of business, claim complexity, and geographic location.

Phase 2 — Acknowledgment: Insurer acknowledges receipt within the timeframe required by state law (typically 10-15 business days). Requests any additional documentation needed.

Phase 3 — Investigation: Adjuster investigates the facts of the loss: inspects damage, interviews parties, reviews police/fire reports, consults medical records (health/liability claims), and assesses coverage.

Phase 4 — Coverage Determination: Adjuster determines whether the loss is covered under the policy. Issues Reservation of Rights letter if coverage is uncertain while investigation continues.

Phase 5 — Evaluation: Adjuster quantifies the loss: obtains repair estimates, reviews medical bills and records, assesses liability, and calculates damages.

Phase 6 — Resolution: Claim settled through payment, denial, or negotiation. Denial must be communicated in writing with specific policy language cited.

Phase 7 — File Closure: Once payment is made and all subrogation/salvage potential is pursued, file is closed.

8.3 Types of Adjusters

Adjuster Type	Employed By	Role
Staff Adjuster	Insurance company (W-2 employee)	Handles claims for one insurer; most common
Independent Adjuster (IA)	Third-party adjusting firm	Contracted by insurers for surge capacity or specialization
Public Adjuster	Policyholder (hired by insured)	Advocates for policyholder; paid % of settlement
Catastrophe (CAT) Adjuster	Insurer or IA firm	Deployed to disaster zones; handles high volume

8.4 Claim Documentation Requirements by Type

Property Claims

- Completed claim form
- Photos and video of all damage
- Home inventory / personal property list with values
- Repair estimates from licensed contractors
- Receipts for emergency repairs / protective measures
- Police report (for theft, vandalism)
- Fire department report (for fire losses)

Auto Claims

- Police accident report
- Photos of all vehicles involved and accident scene
- Contact and insurance information for all parties
- Witness names and contact information
- Medical records and bills (for injury claims)
- Vehicle repair estimates from body shops
- Rental car receipts

Health Claims (When Insured Files)

- Itemized bill from provider (not just summary bill)
- Explanation of Benefits (EOB) from primary insurer (for coordination of benefits)
- Referral authorization if required
- Medical records supporting necessity of treatment

Life Insurance Claims

- Certified death certificate (not a copy)
- Completed claimant's statement

- Original policy document (if available)
- Proof of relationship for non-spouse beneficiaries
- Attending physician's statement (for contestable claims)

8.5 State-Mandated Claim Handling Timelines

Requirement	Typical Timeframe
Acknowledge receipt of claim	10-15 business days
Accept or deny claim after proof of loss	15-40 business days (varies by state)
Pay accepted claim after agreement	5-15 business days
Health claim payment (clean claim)	30-45 days (some states 15-30)
Respond to appeal	30-60 days (60 days for urgent/external)

Note: California: 15 days to acknowledge; 40 days to accept/deny after proof of loss. Texas: 15 days to acknowledge; 15 business days to pay after acceptance. Timelines vary significantly by state.

8.6 Claim Denials — Common Reasons

- Policy lapsed due to non-payment of premium
- Loss falls under a specific exclusion
- Claim filed after the reporting deadline (late notice)
- Insured failed to cooperate with investigation
- Material misrepresentation on the application
- Loss pre-dates policy effective date
- Coverage limits already exhausted
- Insured lacks insurable interest
- Loss caused by intentional act of the insured

8.7 Subrogation

After paying a claim, the insurer may pursue subrogation — recovery of claim costs from a responsible third party. The insured must cooperate with subrogation efforts and cannot settle with the responsible party in a way that prejudices the insurer's subrogation rights.

Example: Your insurer pays \$15,000 for your car damage caused by another driver. Your insurer then pursues the at-fault driver or their insurer for \$15,000 plus any expenses. If your deductible was

\$1,000, you may receive \$1,000 back from the subrogation recovery.

8.8 Appeals and Dispute Resolution

If a claim is denied or the settlement is disputed, the insured has several options:

- **Internal Appeal:** Request insurer review the decision with additional documentation or arguments.
- **State Insurance Department Complaint:** File a complaint with the state regulator. Insurers must respond and justify their decisions.
- **Appraisal:** Property policies often include an appraisal clause — each party hires an appraiser and they select an umpire to resolve disputes about the amount of loss.
- **Mediation:** Neutral third party facilitates settlement discussion. Non-binding.
- **Arbitration:** Binding or non-binding depending on policy; faster and less expensive than litigation.
- **Litigation:** Lawsuit against the insurer for breach of contract or bad faith.

Chapter 9: Insurance Terminology Glossary

Term	Definition
Accident	An unforeseen, sudden, unintended event causing loss or injury
Actual Cash Value (ACV)	Fair market value of property considering age and depreciation
Adjuster	Professional who investigates and settles insurance claims
Agreed Value	Policy pays stated agreed amount without depreciation (fine art, classic cars)
All-Risk	Coverage for all perils not specifically excluded
Appraisal	Formal valuation of property or claim amount by expert
Arbitration	Binding dispute resolution outside of court
Assignment	Transfer of policy rights or benefits to another party
Bad Faith	Insurer's unreasonable or dishonest claims handling
Binder	Temporary evidence of insurance coverage pending policy issuance
Blanket Coverage	Single coverage limit for multiple items or locations
Cancellation	Termination of policy before expiration date
Capacity	An insurer's ability to write new business (financial strength)
Certificate of Insurance	Document proving insurance coverage exists
Claim	Formal request to insurer for payment under the policy terms
Claimant	Party making a claim; can be insured or third party
Coinsurance	Shared cost between insured and insurer; or minimum coverage requirement
Collision	Accident where vehicle strikes another vehicle or object
Comprehensive	Coverage for non-collision vehicle damage
Concealment	Deliberate withholding of material information from insurer
Conditions	Policy provisions defining insured's obligations
Coverage Gap	Period without insurance coverage
Declarations Page	Summary page of policy terms, limits, and parties
Deductible	Amount insured pays per claim before coverage kicks in
Depreciation	Reduction in value due to age, wear, and obsolescence
Dividend	Return of premium to policyholder (mutual company surplus)

Term	Definition
Dwelling	Primary living structure covered under homeowners policy
Endorsement	Written amendment to policy changing its terms
Exclusion	Specific perils, situations, or losses not covered
Expiration Date	Date on which the policy coverage ends
Exposure	Measure of risk subject to loss (e.g., payroll, revenue, vehicles)
First Party	Insured filing claim against their own policy
Floater	Endorsement providing broader or scheduled item coverage
Fraud	Intentional misrepresentation to obtain insurance payment
Hazard	Condition increasing chance or severity of loss
HIPAA	Federal law protecting privacy of health information
Indemnity	Compensation restoring insured to pre-loss financial position
Inland Marine	Coverage for goods in transit and movable property
Insurable Interest	Financial stake in insured person or property
ISO	Insurance Services Office — develops standard policy forms and rating data
Lapse	Termination of policy due to non-payment of premium
Liability	Legal obligation to pay damages caused to others
Limits	Maximum amount insurer will pay under the policy
Loss	Reduction in value or damage resulting from a covered peril
Loss Ratio	Claims paid divided by premiums earned (insurer efficiency metric)
Material Fact	Information that would affect insurer's decision to insure or price risk
Moral Hazard	Increased risk-taking due to having insurance
Named Insured	Person or entity listed on the policy declarations
Negligence	Failure to exercise reasonable care resulting in harm to others
Non-Renewal	Insurer's decision not to renew policy at expiration
Occurrence	Event giving rise to a covered claim
Peril	Cause of loss (e.g., fire, theft, flood)
Policy Period	Dates during which coverage is in effect

Term	Definition
Premium	Price paid for insurance coverage
Pro Rata Cancellation	Refund proportional to unused coverage period
Proof of Loss	Formal statement documenting claim details submitted to insurer
Proximate Cause	Primary cause directly producing the loss
Reinstatement	Restoring lapsed policy, usually after late premium payment
Replacement Cost	Cost to replace property with new item of like kind and quality
Reserve	Funds set aside by insurer to pay anticipated claims
Rider	Additional coverage attached to a policy
Risk	Uncertainty of financial loss; chance of loss occurring
Salvage	Property with residual value after total loss settlement
Schedule	List of specifically insured items with individual values
Self-Insured Retention (SIR)	Amount paid by insured before liability policy responds
Short Rate Cancellation	Insured cancels early; receives less than pro rata refund
Subrogation	Insurer's right to recover payments from responsible third party
Surplus Lines	Non-standard insurance for risks standard market won't cover
Third Party	Party other than insured or insurer involved in a claim
Umbrella Policy	Excess coverage providing additional limits above underlying policies
Underwriter	Professional who evaluates risk and determines coverage terms
Vacancy	Property unoccupied for extended period; may affect coverage
Valued Policy	Pays agreed amount without depreciation upon total loss
Waiver	Voluntary relinquishment of a known right by insurer or insured
Workers' Compensation	Employer-paid coverage for work-related injury/illness

Chapter 10: Regulatory & Compliance Framework

10.1 State Regulation

In the United States, insurance is primarily regulated at the state level. Each state has an Insurance Commissioner or Department of Insurance (DOI) responsible for licensing insurers and agents, approving rates and policy forms, investigating complaints, and enforcing consumer protection laws.

10.2 Key Federal Laws Affecting Insurance

Affordable Care Act (ACA) — 2010

Requires health insurers to cover essential health benefits, prohibits denial for pre-existing conditions, allows children to stay on parents' plan until age 26, eliminates lifetime and annual dollar limits on essential benefits, requires coverage of preventive services without cost-sharing.

HIPAA — Health Insurance Portability and Accountability Act

Protects the privacy and security of individuals' health information. Insurers and their business associates must implement administrative, physical, and technical safeguards. Violations can result in civil and criminal penalties up to \$1.9M per category violation annually.

ERISA — Employee Retirement Income Security Act

Governs employer-sponsored benefit plans including group health, disability, and life insurance. Preempts most state insurance laws for self-funded employer plans. Sets minimum standards for plan administration, reporting, and fiduciary responsibility.

McCarran-Ferguson Act — 1945

Grants states the right to regulate insurance. Provides antitrust exemption for insurers to share data (loss statistics) necessary for rate-making. Congress can override with specific legislation.

10.3 State Insurance Guaranty Funds

Every state has a guaranty association that protects policyholders if their insurer becomes insolvent. Coverage limits vary by state and line of business:

Coverage Type	Typical Guaranty Fund Limit
Life insurance death benefit	\$300,000 per policy
Health insurance claims	\$500,000 per policy

Coverage Type	Typical Guaranty Fund Limit
Annuity cash value	\$250,000 per policy
Property/Casualty claims	\$300,000-\$500,000 per claim

Note: Guaranty funds are funded by assessments on licensed insurers — not taxpayer funds.

10.4 Key Compliance Requirements for Claim Handling

- Acknowledge claims within state-mandated timeframes
- Conduct thorough, prompt, and impartial investigation
- Communicate coverage decisions with specific policy language cited
- Not deny claims without reasonable basis (bad faith standard)
- Pay undisputed amounts promptly while disputed portions are resolved
- Maintain complete claim files for statutory retention periods (typically 6-10 years)
- Protect confidential policyholder and claimant information per privacy laws
- Follow state-specific requirements for earthquake, flood, and hurricane claims

Chapter 11: Fraud Detection & Prevention

11.1 The Cost of Insurance Fraud

Insurance fraud costs the U.S. insurance industry an estimated \$308 billion annually (Coalition Against Insurance Fraud, 2022). These costs are passed on to policyholders through higher premiums — approximately \$900 per year per U.S. household. Fraud occurs across all lines of insurance.

11.2 Types of Insurance Fraud

Hard Fraud

Deliberate staging or fabrication of a loss. Examples: staging car accidents ('swoop and squat'), arson, faking theft, staging slip-and-fall accidents, filing claims for non-existent damage.

Soft Fraud (Opportunistic Fraud)

Exaggerating legitimate claims. Examples: inflating repair bills, including pre-existing damage in a new claim, misrepresenting facts on an application to get lower premiums, padding medical bills after a real accident.

Application Fraud

Misrepresenting information when applying for insurance to obtain coverage or lower premiums. Examples: hiding DUI history on auto application, not disclosing pre-existing conditions, misrepresenting property use.

11.3 Red Flags by Insurance Type

Auto Insurance

- Accident reported days after it allegedly occurred
- Multiple claimants with no independent witnesses
- Injuries inconsistent with low-speed accident damage
- Claimant immediately hires attorney and medical provider
- Vehicle had prior damage in the area of claimed damage
- Policy purchased immediately before reported accident
- Claimant demands cash settlement, refuses inspection

Property Insurance

- Policy recently increased right before a loss

- Financial difficulties: recent foreclosure, bankruptcy filings
- Valuables conveniently absent during theft/fire
- No evidence of forced entry for burglary claim
- Claim amount exactly equals deductible threshold for upgrade
- Contractor and insured have prior relationship

Health Insurance

- Billing for services not rendered
- Upcoding: billing higher-cost procedure than performed
- Unbundling: billing separately for procedures typically bundled
- Unusual frequency of services from single provider
- Identical claim forms with varying patient information

11.4 Special Investigative Units (SIU)

Insurers maintain SIUs staffed with fraud investigators, often former law enforcement. When claims meet fraud referral criteria, the claim is referred to SIU for enhanced investigation. Techniques include: recorded statement analysis, scene investigation, social media surveillance, database searches (ISO ClaimSearch, NICB), medical record review, and coordination with law enforcement.

11.5 Fraud Reporting

Suspected fraud should be reported to:

- The insurer's SIU or fraud hotline
- State Insurance Fraud Bureau or DOI fraud division
- National Insurance Crime Bureau (NICB): 1-800-TEL-NICB
- FBI (for large-scale healthcare fraud schemes)

Note: Reporting suspected fraud is protected in most states. Insurers acting in good faith are immune from defamation claims when reporting fraud to authorities.

Chapter 12: Customer Service Guidelines for Claims Support

12.1 Core Service Principles

- **Empathy First:** Claimants have often experienced traumatic losses. Acknowledge the emotional impact before addressing logistics.
- **Clarity:** Avoid insurance jargon. Explain coverage decisions in plain language with specific examples.
- **Transparency:** Be honest about timelines, processes, and coverage limitations. Manage expectations proactively.
- **Timeliness:** Return calls within 24 hours. Provide regular status updates even when there is no new information.
- **Documentation:** Summarize all conversations in the claim file. Follow up verbal commitments with written confirmation.

12.2 Difficult Conversation Frameworks

Delivering a Coverage Denial

1. Express empathy for the situation. 2. State the decision clearly and early in the conversation. 3. Cite the specific policy language and exclusion. 4. Explain the reason the exclusion applies to this loss. 5. Outline appeal rights and process. 6. Provide contact information for state insurance department.

Handling Disputes About Settlement Amount

1. Acknowledge the claimant's perspective. 2. Walk through how the settlement was calculated step by step. 3. Explain depreciation methodology if ACV policy. 4. Offer to review any additional documentation the claimant can provide. 5. Explain the appraisal or dispute resolution process available under the policy.

12.3 Service Level Standards

Activity	Target Response Time
FNOL acknowledgment	Within 1 business day
Return missed calls	Within 4 business hours
Adjuster assignment notification	Within 2 business days of FNOL

Activity	Target Response Time
Field inspection scheduling	Within 5 business days
Coverage determination letter	Within 15 business days of receiving complete information
Payment processing after agreement	Within 5 business days
Appeal response	Within 30 calendar days
Customer complaint escalation response	Within 24 business hours

12.4 Common Customer Questions & Answers

'Why is my claim taking so long?'

Explain the investigation steps required: inspections, records gathering, liability determination, and any third-party involvement. Provide a specific next step and timeline. Log a follow-up date in the claim file.

'My settlement offer is too low — what can I do?'

Advise the claimant they can provide additional documentation (additional repair estimates, receipts, appraisals). Explain the policy's appraisal clause if applicable. Mention the option to consult a public adjuster.

'Will filing a claim raise my rates?'

Rate increases depend on the insurer, the claim type, and state regulations. At-fault auto claims and weather claims may be treated differently. Claims-free discounts may be lost. Recommend reviewing the policy and consulting with their agent.

'My policy was cancelled — what do I do?'

First confirm the reason: non-payment vs. underwriting cancellation. If non-payment, advise on reinstatement options. If underwriting, advise on obtaining coverage through the assigned risk pool or surplus lines market. Provide the state DOI contact for assistance.

12.5 Regulatory Compliance in Customer Interactions

- Never promise coverage before investigation is complete
- Never make statements that could be construed as waiving coverage defenses
- Provide written claim status updates as required by state law
- Use only state-approved denial language and forms
- Maintain recordings of customer calls per state requirements
- Comply with Do Not Call lists and communication preference laws

— End of Insurance Knowledge Base Document —

This document was prepared for use as RAG ingestion data for the Insurance Claim Support AI Agent project. It covers the major insurance lines, claims processes, regulatory environment, and customer service best practices necessary to power an intelligent, context-aware insurance support copilot.

Document Version: 1.0 | Prepared for: LangMem + RAG AI Agent Project | Total Pages: ~50

Appendix A: Coverage Quick Reference Matrix

A.1 Homeowners Insurance Peril Coverage by Policy Form

Peril / Cause of Loss	HO-1	HO-2	HO-3 Dwelling	HO-3 Contents	HO-5
Fire & Lightning	Yes	Yes	Yes	Yes	Yes
Windstorm & Hail	Yes	Yes	Yes	Yes	Yes
Explosion	Yes	Yes	Yes	Yes	Yes
Riot or Civil Commotion	Yes	Yes	Yes	Yes	Yes
Aircraft Damage	Yes	Yes	Yes	Yes	Yes
Vehicle Damage	Yes	Yes	Yes	Yes	Yes
Smoke Damage	Yes	Yes	Yes	Yes	Yes
Vandalism	No	Yes	Yes	Yes	Yes
Theft	No	Yes	Yes	Yes	Yes
Falling Objects	No	Yes	Yes	Yes	Yes
Weight of Ice/Snow/Sleet	No	Yes	Yes	Yes	Yes
Accidental Discharge (Plumbing)	No	Yes	Yes	Yes	Yes
Freezing of Household Systems	No	Yes	Yes	Yes	Yes
Electrical Surge Damage	No	Yes	Yes	Yes	Yes
Sudden Cracking of Water Heater	No	Yes	Yes	Yes	Yes
All Other Perils (Open)	No	No	Yes	No	Yes
Flood	No	No	No	No	No
Earthquake	No	No	No	No	No
Wear & Tear / Neglect	No	No	No	No	No

A.2 Auto Insurance Coverage Applicability

Scenario	Liability	Collision	Comprehensive	UM/UIM	PIP/MedPay
You hit another car (at fault)	Pays other's damages	Pays your car	No	No	Pays your medical
Other driver hits you (at fault)	No	Pays your car*	No	Pays if other uninsured	Pays your medical

Scenario	Liability	Collision	Comprehensive	UM/UIM	PIP/MedPay
You hit a deer	No	No	Yes	No	No
Your car is stolen	No	No	Yes	No	No
Hail damages your car	No	No	Yes	No	No
You hit a pedestrian	Pays pedestrian	No	No	No	No
Passenger injured in your car	No	No	No	No	Yes
You damage your garage	No	No	No	No	No

*Note: *Collision pays minus deductible; you would seek reimbursement from at-fault driver or their insurer.*

A.3 Life Insurance Product Comparison Matrix

Feature	Term Life	Whole Life	Universal Life	Variable Life
Coverage Duration	Set term (10-30 yr)	Lifetime	Lifetime	Lifetime
Premium	Lowest	Highest (fixed)	Flexible	Variable
Cash Value	None	Guaranteed growth	Interest-linked	Market-linked
Death Benefit	Level or decreasing	Guaranteed	Flexible	Variable
Investment Risk	None	None	Low (guaranteed min)	High (policyholder)
Loan Against Policy	No	Yes	Yes	Yes
Best For	Pure protection	Estate planning	Flexibility	Growth potential
Complexity	Simple	Moderate	Moderate-High	High

Appendix B: Claim Forms, Medical Codes & Industry Standards

B.1 Standard Claim Forms

Form	Used For	Submitted By
CMS-1500	Professional (physician) health insurance claims	Healthcare providers
UB-04 (CMS-1450)	Institutional claims (hospital, facility)	Hospitals, facilities
AIA Property Loss Notice	Property insurance claims	Policyholder/agent
ACORD 1	Property loss notice	Agent to insurer
ACORD 2	Auto/vehicle loss notice	Agent to insurer
ACORD 3	Inland marine loss notice	Agent to insurer
SR-1 / SR-21	State accident report (auto)	Drivers/insurers to DMV
Proof of Loss (POL)	Formal claim statement under oath	Policyholder to insurer

B.2 Medical Coding Systems for Health Claims

ICD-10-CM — Diagnosis Codes

The International Classification of Diseases, 10th Revision (ICD-10-CM) is used to document diagnoses on all healthcare claims. ICD-10 codes consist of 3-7 alphanumeric characters (e.g., J18.9 = Unspecified pneumonia; S72.001A = Fracture of right femoral neck, initial encounter). The U.S. transitioned from ICD-9 to ICD-10 on October 1, 2015.

CPT — Current Procedural Terminology

CPT codes (maintained by the AMA) are 5-digit numeric codes describing medical services performed. Categories: Category I (clinical procedures), Category II (performance measurement), Category III (emerging technologies). Common examples: 99213 = Office visit established patient moderate complexity; 71046 = Chest X-ray 2 views; 93000 = Electrocardiogram routine.

HCPCS Level II

Healthcare Common Procedure Coding System Level II codes cover supplies, equipment, and services not described by CPT. Alphanumeric format (letter + 4 digits). Used for Medicare/Medicaid and some commercial claims. Examples: A4253 = Blood glucose test strips; E0601 = CPAP device; J0696 = Ceftriaxone injection.

DRG — Diagnosis Related Groups

Medicare uses DRGs to pay hospitals a fixed amount per admission based on diagnosis and procedure, regardless of actual cost. Over 750 DRGs exist. Hospitals are incentivized to discharge patients efficiently and manage costs.

B.3 Liability Claims — Legal Standards

Negligence Elements (Must Prove All Four)

- **Duty:** Defendant owed a duty of care to the plaintiff
- **Breach:** Defendant breached that duty through action or inaction
- **Causation:** Breach directly caused the plaintiff's harm (actual and proximate cause)
- **Damages:** Plaintiff suffered quantifiable harm as a result

Comparative vs. Contributory Negligence

Most states use comparative negligence — damages are reduced by the plaintiff's percentage of fault. Pure comparative: recovery allowed even if 99% at fault. Modified comparative: recovery barred if 50% or 51% at fault (varies by state). A few states still use contributory negligence — any fault by plaintiff bars all recovery.

B.4 Property Valuation Methods

Method	Definition	Used For
Actual Cash Value (ACV)	Replacement cost minus physical depreciation	Standard property coverage
Replacement Cost Value (RCV)	Cost to replace with new item of like kind and quality	Upgraded property coverage
Functional Replacement Cost	Cost of a functionally equivalent modern replacement	Replacements, obsolete features
Agreed Value	Pre-agreed amount; no depreciation applied	Fine art, classic cars, unique property
Market Value	What property would sell for (not replacement)	Real estate transactions (not typically insurance)
Stated Amount	Maximum insured pays; may depreciate	Some state-chartered inland marine

Appendix C: State Insurance Reference Data

C.1 No-Fault vs. Tort States (Auto Insurance)

In no-fault states, each driver's own PIP coverage pays for their medical expenses after an accident regardless of fault. Tort (at-fault) states allow injured parties to sue the at-fault driver for full damages. No-fault states generally require PIP coverage and restrict the right to sue.

State Type	States (Sample)	Key Feature
Pure No-Fault	Michigan, New Jersey (choice), Pennsylvania (choice)	Claims for general damages below threshold
No-Fault with Verbal Threshold	New York, Florida, New Jersey (default)	Lawsuit allowed only for serious injury (death, significant disfigurement)
No-Fault with Monetary Threshold	Minnesota, Massachusetts, Hawaii	Lawsuit allowed when medical expenses exceed threshold
Tort (At-Fault) States	California, Texas, Illinois, Ohio, and majority	Injured party sues at-fault driver; no required PIP
Choice No-Fault	Pennsylvania, New Jersey, Kentucky	Drivers select no-fault or tort system at purchase

C.2 Flood Insurance — NFIP Key Facts

Standard homeowners and renters policies DO NOT cover flood damage. Separate flood insurance must be obtained through the National Flood Insurance Program (NFIP, administered by FEMA) or private flood insurers.

- NFIP maximum coverage: \$250,000 for building; \$100,000 for contents
- 30-day waiting period for new policies (exceptions for loan closing and map revisions)
- Flood zones: Zone A and AE = high risk (1% annual chance); Zone X = low-moderate risk
- Properties in high-risk zones with federally-backed mortgages are REQUIRED to carry flood insurance
- Average NFIP premium: approximately \$700-\$900 per year (varies widely by flood zone and elevation)
- Private flood insurance may offer higher limits and broader coverage than NFIP

C.3 Earthquake Insurance

Standard homeowners policies exclude earthquake damage. Earthquake coverage is available as an endorsement or separate policy. In California, the California Earthquake Authority (CEA) is the primary source.

Feature	Detail
Deductible	Typically 10-25% of dwelling coverage (not a flat dollar amount)
Coverage	Dwelling repair, personal property, ALE, building code upgrades
Common Exclusions	Pre-existing damage, flood/tsunami following quake, vehicles
High-Risk States	California, Alaska, Hawaii, Washington, Oregon, Missouri (New Madrid fault)
CEA Policy	California Earthquake Authority — largest residential earthquake insurer in U.S.

C.4 Key Insurance Regulatory Bodies

Organization	Role
NAIC — National Association of Insurance Commissioners	Coordinates state regulators; develops model laws and consumer tools
State Departments of Insurance (DOI)	License insurers/agents, approve rates/forms, handle complaints
FEMA / NFIP	Administers national flood insurance program
CMS — Centers for Medicare & Medicaid Services	Oversees federal health insurance programs and ACA compliance
DOL — Department of Labor	Regulates employer benefit plans under ERISA
FIO — Federal Insurance Office (Treasury)	Monitors industry; coordinates international policy; no regulatory power
NCUA	Regulates credit union-sponsored insurance products
SEC	Oversees variable life and annuity products (securities component)

Appendix D: AI Agent Q&A; Scenario Examples

This appendix provides representative question-and-answer scenarios illustrating how the Insurance Claim Support AI Agent should respond using the knowledge in this corpus.

D.1 Health Insurance Scenarios

Scenario 1: Pre-authorization Denial

User: My doctor ordered an MRI but my insurance denied it. What can I do?

Agent Response Guidance: Explain that this is a pre-authorization (prior authorization) denial. The insurer has determined the service does not meet medical necessity criteria at this time. Steps: (1) Ask the doctor's office to submit additional clinical documentation supporting medical necessity. (2) Request a peer-to-peer review between the treating physician and the insurer's medical director. (3) If still denied, file a formal internal appeal within the timeframe specified in the denial letter (typically 30-60 days). (4) If internal appeal fails, request an Independent Medical Review (IMR)/external appeal. Inform the user of their right to appeal under ACA rules.

Scenario 2: Out-of-Network Emergency

User: I had an emergency and was taken to an out-of-network hospital. Will my HMO cover it?

Agent Response Guidance: Under ACA rules and the No Surprises Act (2022), emergency services must be covered by HMOs even at out-of-network facilities. The insurer must apply in-network cost-sharing (deductible, copay, coinsurance) for emergency services. The provider cannot balance-bill beyond the allowed amount. Advise the user to submit the claim and, if denied for being out-of-network, appeal citing the emergency exception and No Surprises Act protections.

D.2 Auto Insurance Scenarios

Scenario 3: Total Loss Dispute

User: My insurer declared my car a total loss and offered \$12,000, but I owe \$16,000 on my loan. Help!

Agent Response Guidance: This is a Gap insurance situation. Standard auto insurance pays ACV, not the loan balance. First, question whether the ACV calculation is accurate — request the valuation report and comparable vehicle sales data. If the ACV is fair but there is still a shortfall, Gap insurance (if purchased) covers the difference. If no Gap insurance, the owner remains responsible for the \$4,000 difference to the lender. Advise the user to review their original loan documents for Gap coverage inclusion and check with their dealer/lender.

Scenario 4: Rental Coverage After Accident

User: My car is in the shop after an accident. The repairs are taking 3 weeks. Will insurance pay for my rental?

Agent Response Guidance: Coverage depends on (1) whether the insured purchased rental reimbursement coverage on their own policy, and (2) fault. If the other driver was at fault, their liability insurer should pay for the rental (check state law for rental duration limits). If the insured was at fault and has rental reimbursement, their own policy covers it up to the daily and maximum policy limits (e.g., \$30/day up to \$900). Advise the user to confirm their specific limits and keep all rental receipts.

D.3 Property Insurance Scenarios

Scenario 5: Water Damage — Covered or Not?

User: My basement flooded after heavy rain. Will my homeowners insurance cover the damage?

Agent Response Guidance: This depends on the source of the water. (1) If rainwater entered through a window, door, or foundation crack — this is likely flood/surface water, which is EXCLUDED from standard homeowners policies. A separate flood policy (NFIP or private) would be needed. (2) If a sump pump failed and caused backup — sewer/drain backup coverage may apply IF the insured added this endorsement. (3) If a pipe burst and caused the flooding — this is typically covered under the standard policy's plumbing water damage peril. Advise the user to document the water source carefully for the adjuster's investigation.

Scenario 6: Jewelry Claim

User: My diamond engagement ring was stolen from a hotel room. Is it covered under my renters insurance?

Agent Response Guidance: Renters insurance covers theft of personal property, but jewelry typically has a sub-limit (e.g., \$1,500 for all jewelry theft combined). If the ring is worth more than the sub-limit, the insured would only receive up to \$1,500 minus their deductible. To get full coverage for high-value jewelry, a scheduled personal property endorsement (floater) should have been purchased, which would insure the specific item for its appraised value with a lower or no deductible. Advise the user to file the claim, submit police report and jewelry purchase receipt/appraisal, and expect payment up to the jewelry sub-limit.

D.4 Life Insurance Scenarios

Scenario 7: Claim During Contestability Period

User: My husband passed away 8 months after purchasing a life insurance policy. The insurer is investigating. Is this normal?

Agent Response Guidance: Yes, this is standard and legal. Life insurance policies have a 2-year contestability period during which the insurer can review the original application for material misrepresentation or fraud. If the insured answered all application questions honestly, the claim will be paid. The insurer typically requests: original application, medical records, and an Attending Physician's Statement. This process usually takes 60-90 days. Advise the beneficiary to cooperate fully with the investigation, provide requested records promptly, and consult an attorney if the claim is improperly denied.

D.5 Claims Process Scenarios

Scenario 8: Bad Faith Complaint

User: My insurer has been 'investigating' my property claim for 6 months with no resolution. What are my rights?

Agent Response Guidance: Most states require insurers to accept or deny claims within 40 days of receiving a complete proof of loss. A 6-month investigation without resolution may constitute unreasonable delay — a form of insurance bad faith. Steps: (1) Send a formal written demand to the insurer citing the state's prompt payment statute. (2) File a complaint with the state Department of Insurance (DOI) — insurers must respond to DOI complaints. (3) Consult an insurance bad faith attorney who may work on contingency. (4) Document all communications, delays, and contacts with the insurer as evidence of bad faith.

End of Document — All Appendices Complete

This 50-page Insurance Knowledge Base is structured for optimal RAG ingestion. Each chapter and section contains dense, factual, domain-specific content suitable for chunking and embedding. The document covers terminology, procedures, regulatory requirements, and real-world claim scenarios to ensure the AI agent can provide accurate, context-aware insurance support.

Appendix E: Insurance Industry Statistics & Benchmarks

E.1 U.S. Auto Insurance Benchmarks

Metric	Benchmark / Average
Average U.S. auto insurance premium (2023)	\$2,014/year (full coverage); \$622 (liability only)
Average bodily injury claim severity	\$24,000-\$30,000
Average property damage claim severity	\$4,500-\$5,500
Average comprehensive claim	\$1,800-\$2,200
Average collision claim	\$3,800-\$4,200
Claims frequency (BI per 100 insured vehicles)	0.8-1.2 claims per year
Auto theft rate (U.S.)	Approximately 721,000 vehicles stolen annually
Total loss threshold (typical)	70-80% of ACV
Days to close a simple auto claim	7-15 business days
Days to close a complex injury claim	60-365+ days

E.2 Homeowners Insurance Benchmarks

Metric	Benchmark / Average
Average U.S. homeowners premium (2023)	\$1,428/year (varies dramatically by state)
Most expensive state for HO insurance	Oklahoma (~\$5,000/year), followed by Kansas, Nebraska
Least expensive states	Hawaii, Vermont, Delaware (~\$600-\$800/year)
Average fire claim	\$75,000-\$100,000
Average water damage claim (non-flood)	\$12,000-\$15,000
Average wind/hail claim	\$11,000-\$15,000
Average theft claim (HO)	\$4,000-\$5,000
Percentage of homes with flood insurance	Approximately 4% nationally
Average time to settle property CAT claim	30-90 days

E.3 Health Insurance Benchmarks

Metric	Benchmark / Average
Average employer-sponsored plan premium (2023)	\$7,911/year (individual); \$22,463/year (family)
Average employer contribution	~83% of individual; ~73% of family premium
Average individual deductible (employer plan)	\$1,735
Average out-of-pocket maximum	\$4,272 individual; \$8,544 family
ACA marketplace average premium (2023)	\$438/month before subsidies
Medicare enrollment	65+ million Americans enrolled
Medicaid/CHIP enrollment	93+ million Americans
Clean claim payment time	14-21 days for electronic claims
Health insurance denial rate (initial)	Approximately 17% of in-network claims denied initially

E.4 Escalation Guide — When AI Must Transfer to Human

Scenario	Priority	Reason
Claimant in emotional crisis	IMMEDIATE	Safety; human empathy required
Active emergency (fire, accident scene, etc.)	IMMEDIATE	Direct to 911; not AI's role
Death-related claim	HIGH	Specialist empathy and handling
Large-loss >\$250K	HIGH	Complex coverage analysis
Suspected fraud indicators	HIGH	SIU referral required
Regulatory complaint or litigation threat	HIGH	Legal/compliance review
Coverage dispute needing legal interpretation	MEDIUM	Attorney review may apply
Claimant alleges bad faith	HIGH	Escalate to management and legal
Media inquiry about a claim	HIGH	Refer to communications

E.5 Key Industry Resources

Resource	Contact
National Flood Insurance Program (NFIP)	1-800-638-6620 floodsmart.gov
NICB Fraud Hotline	1-800-TEL-NICB (1-800-835-6422)

Resource	Contact
Medicare Helpline	1-800-MEDICARE (1-800-633-4227)
Healthcare.gov (ACA Marketplace)	1-800-318-2596
NAIC Consumer Helpline	1-816-783-8500 naic.org
Insurance Information Institute	iii.org
California DOI	1-800-927-4357 insurance.ca.gov
Texas DOI	1-800-252-3439 tdi.texas.gov

Insurance Knowledge Base — Complete

This comprehensive Insurance Knowledge Base has been optimized for RAG ingestion. Recommended chunking strategy: chunk by section (H2 level) with 200-500 token chunks and 50-token overlap. Embed using a domain-appropriate model and store with metadata: chapter, insurance_type, topic, and content_category (definition/process/regulation/statistics/scenario).

Appendix F: Sample Insurance Policy Language

F.1 Standard Insuring Agreement Language (Property)

We will pay for direct physical loss of or damage to Covered Property at the premises described in the Declarations caused by or resulting from any Covered Cause of Loss. Covered Causes of Loss means direct physical loss unless the loss is excluded or limited in this policy.

F.2 Standard Exclusion Language

We will not pay for loss or damage caused directly or indirectly by any of the following. Such loss or damage is excluded regardless of any other cause or event that contributes concurrently or in any sequence to the loss.

- Earth Movement: earthquake, landslide, mine subsidence, earth sinking, rising, or shifting
- Governmental Action: seizure, confiscation, or destruction of property by governmental authority
- Nuclear Hazard: nuclear reaction, radiation, or radioactive contamination
- War and Military Action: warlike action, insurrection, revolution
- Water: flood, surface water, waves, tidal water, overflow of a body of water

F.3 Standard Conditions — Duties After Loss

In case of loss or damage to Covered Property, you must see that the following are done:

- Notify the police if a law may have been broken
- Give us prompt notice of the loss or damage, including a description of the property involved
- As soon as possible, give us a description of how, when, and where the loss or damage occurred
- Take all reasonable steps to protect the Covered Property from further damage
- At our request, give us complete inventories of the damaged and undamaged property
- As often as may be reasonably required, permit us to inspect the property and records
- Send us a signed, sworn proof of loss containing the information we request within 60 days
- Cooperate with us in the investigation or settlement of the claim

F.4 Appraisal Clause — Standard Language

If we and you disagree on the value of the property or the amount of loss, either may make written demand for an appraisal of the loss. In this event, each party will select a competent and impartial

appraiser. The two appraisers will select an umpire. If they cannot agree, either may request that selection be made by a judge of a court having jurisdiction. The appraisers will state separately the value of the property and amount of loss. If they fail to agree, they will submit their differences to the umpire. A decision agreed to by any two will be binding. Each party will pay its own appraiser and bear the other expenses of the appraisal and umpire equally.

F.5 Subrogation Clause — Standard Language

If any person or organization to or for whom we make payment under this Coverage Form has rights to recover damages from another, those rights are transferred to us to the extent of our payment. That person or organization must do everything necessary to secure our rights and must do nothing after loss to impair them. But you may waive your rights against another party in writing prior to a loss to that party.

END OF INSURANCE KNOWLEDGE BASE

Appendix G: RAG Chunking & Ingestion Recommendations

G.1 Recommended Document Chunking Strategy

For optimal RAG performance with this insurance knowledge base, the following chunking strategy is recommended:

- **Chunk Size:** 300-500 tokens per chunk for most sections; 100-200 tokens for glossary entries
- **Overlap:** 50-token overlap between adjacent chunks to preserve context across chunk boundaries
- **Chunk Boundaries:** Prefer H2 section boundaries; never split a table across two chunks
- **Metadata Fields:** chapter_number, chapter_name, section, insurance_type, content_type
- **Content Types to Tag:** definition, process_step, coverage_detail, exclusion, regulation, benchmark, scenario

G.2 Recommended Embedding Models

Model	Provider	Recommended Use Case
text-embedding-3-large	OpenAI	High accuracy; best for production
text-embedding-3-small	OpenAI	Cost-efficient; good for development
embed-english-v3.0	Cohere	Strong retrieval performance
all-mpnet-base-v2	HuggingFace	Open-source; good baseline
bge-large-en-v1.5	BAAI/HuggingFace	Strong open-source option

G.3 Vector Store Options

Vector Store	Type	Best For
FAISS	In-memory/local	Development, small datasets, no infra overhead
Chroma	Local/self-hosted	Development and lightweight production
Pinecone	Managed cloud	Production; high scale; easy integration
Weaviate	Managed/self-hosted	Multi-tenant; rich filtering capabilities
Qdrant	Self-hosted/cloud	High performance; on-premise compliance

Vector Store	Type	Best For
pgvector	PostgreSQL extension	Teams already using PostgreSQL

G.4 LangMem Integration Notes

When integrating this knowledge base with LangMem for the Insurance Claim Support AI Agent, consider the following memory architecture:

- **Semantic Memory:** This RAG corpus serves as the agent's long-term semantic memory — factual knowledge about insurance products, processes, and regulations.
- **Episodic Memory:** LangMem should store per-user interaction history — previous claims, reported incidents, preferences, and prior resolutions — as episodic memory.
- **Working Memory:** Current conversation context, active claim details, and session state maintained in the agent's working memory (LangGraph state).
- **Memory Priority:** Episodic memory (user-specific) should take precedence over semantic memory when they conflict — a user's actual policy details override general information.
- **Retrieval Strategy:** Hybrid retrieval (semantic + keyword BM25) outperforms pure vector search for insurance queries with specific terminology.